From words to a model

Optimization · Level 2 · Lesson 1



Today we will

- Write decisions as variables with units.
- Write limits as inequalities and the goal as one expression.
- Check a plan by putting its numbers in.

Remember

- Variables are decisions; prices and nutrients are data.
- “At least” is ≥, “at most” is ≤; the goal is minimised or maximised.
- A model knows only the rules you write: ask who the answer affects.

Model a class purchase

1. Warm-up (5 min): What do we need to buy this term?
2. Teach (10 min): Variables, constraints, objective.
3. Model (10 min): Write our purchase as a model.
4. Check (10 min): Groups propose plans; we put the numbers in.
5. Wrap-up (5 min): What does the model leave out?

Use small, round prices so pupils can check plans in their heads, for example books at 8 coins, pens at 5 and a budget of 200.

Let’s discuss

1. Why are prices data and not variables?
2. Why does “nearly” not count for a constraint?
3. Which rule did our model forget, and who would notice first?

Exit ticket

- Write one variable, one constraint and the objective for a decision at home.